Development of the Eye



Learning Objectives

- At the end of the lecture, student should be able to know,
- The steps of development of human eye.
- The derivatives of different embryonic primitive eye layers.
- The development of various layers of eye individually, along with optic nerve.

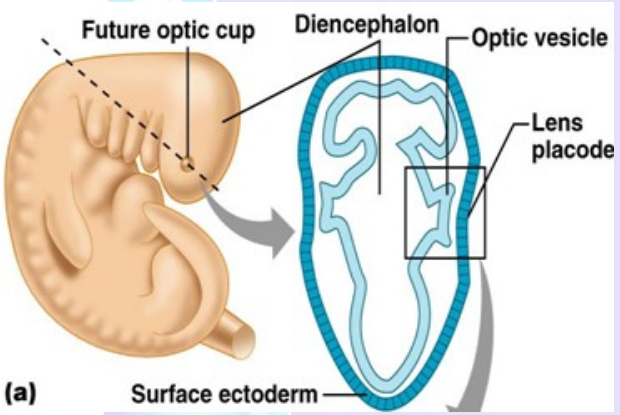
Embryonic Development of the Eye

- Eyes develop as outpocketings of the brain (diencephalon)
- By week 4: Optic vesicles protrude from the diencephalon
- Ectoderm thickens and forms

a lens placodes;

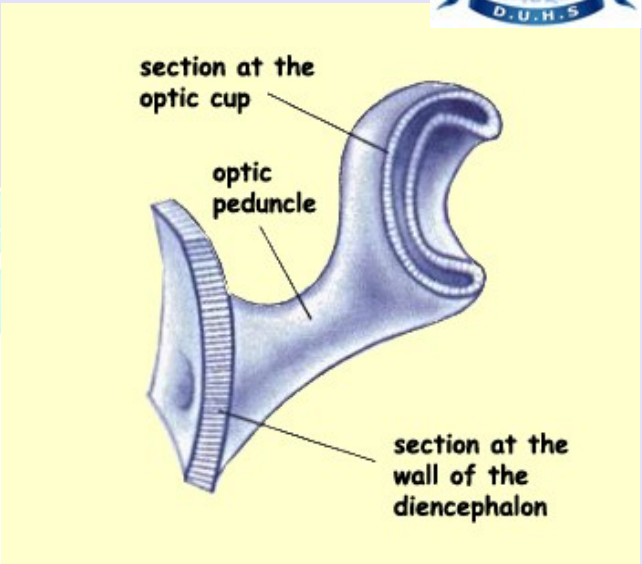
by week 5 – a lens vesicle forms

- <u>Optic cup</u>
  - Internal layer → neural retina
  - External layer → pigmented retina
- <u>Optic fissure</u> – pathway for blood vessels

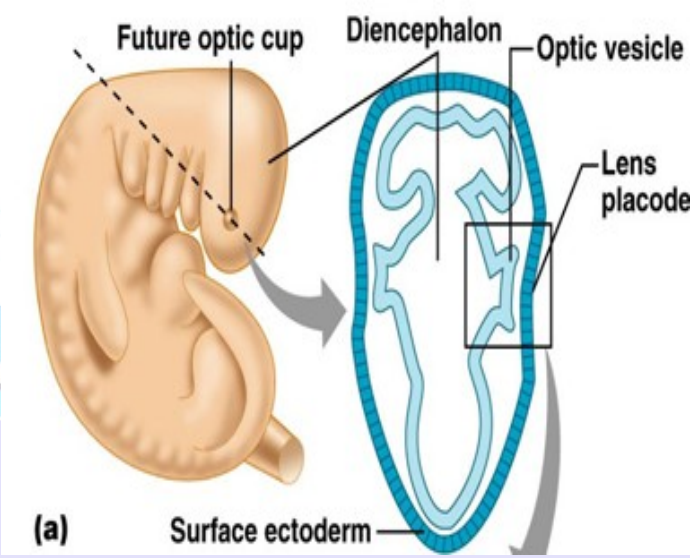


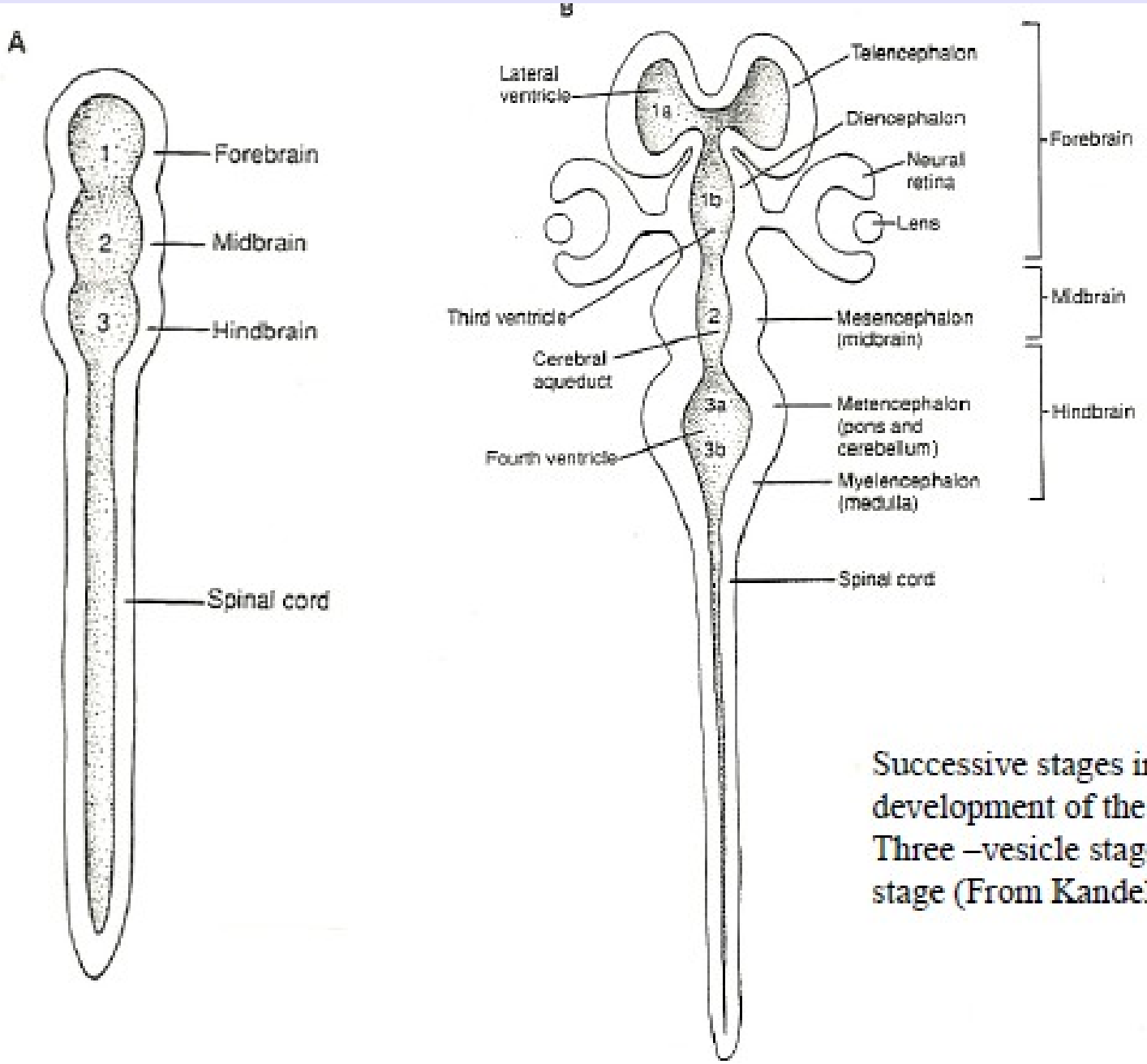
Optic Cup and Lens Vesicle

- The developing eye appears in the 22-day embryo as pair of shallow grooves on the sides of forebrain,these grooves form outpocketings of the forebrain,the **optic vesicles.**



- These vesicles subsequently come in contact with the surface ectoderm and induce changes in the ectoderm necessary for **lens** formation .
- Shortly thereafter the optic vesicle begins to invaginate and forms the double-walled **optic cup.***The inner and outer layers* of this cup are initially separated by a lumen, the intraretinal Space , *but soon this lumen* disappears, and the two layers appose each other.





Successive stages in the development of the neural tube. A: Three –vesicle stage. B: Five vesicle stage (From Kandell and Scwartz)

Embryonic Development of the Eye

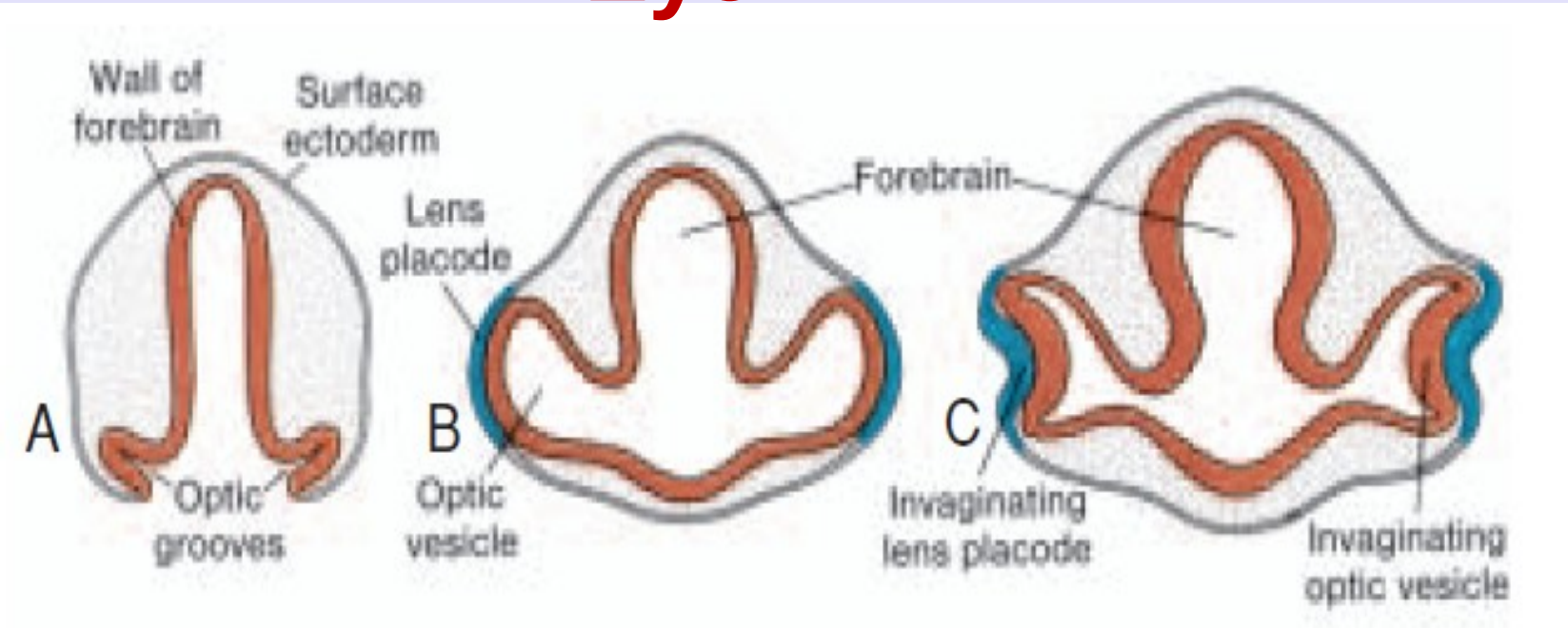
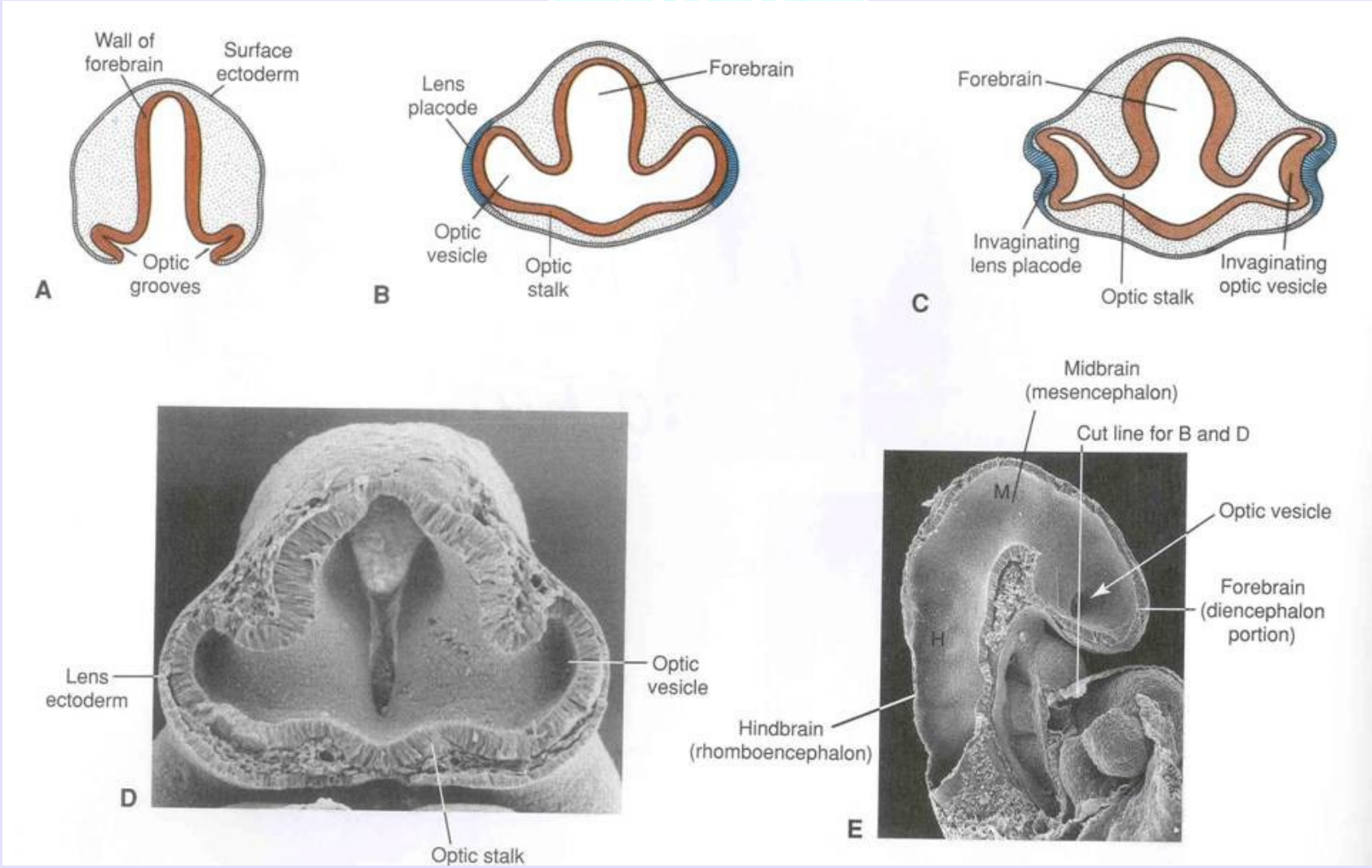


Figure 17.1 A. Transverse section through the forebrain of a 22-day embryo (approximately14 somites), showing the optic grooves
B. Transverse section through the forebrain
of a 4-week embryo showing the optic vesicles in contact with the surface ectoderm. Note the slight thickening of the ectoderm (lens placode).
C. Transverse section
through the forebrain of a 5-mm embryo showing invagination of the optic vesicle and the lens placode.

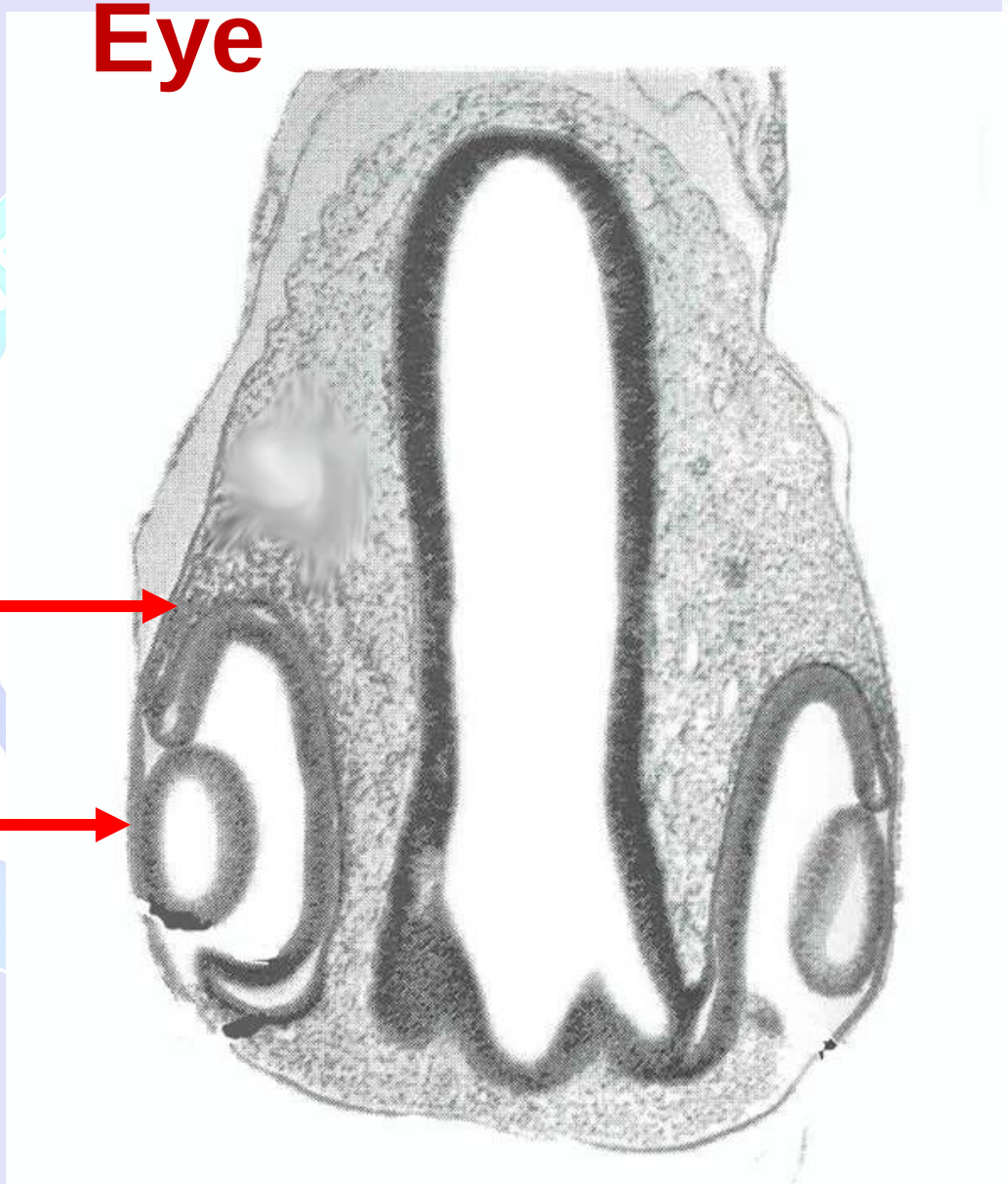
Embryonic Development of the Eye



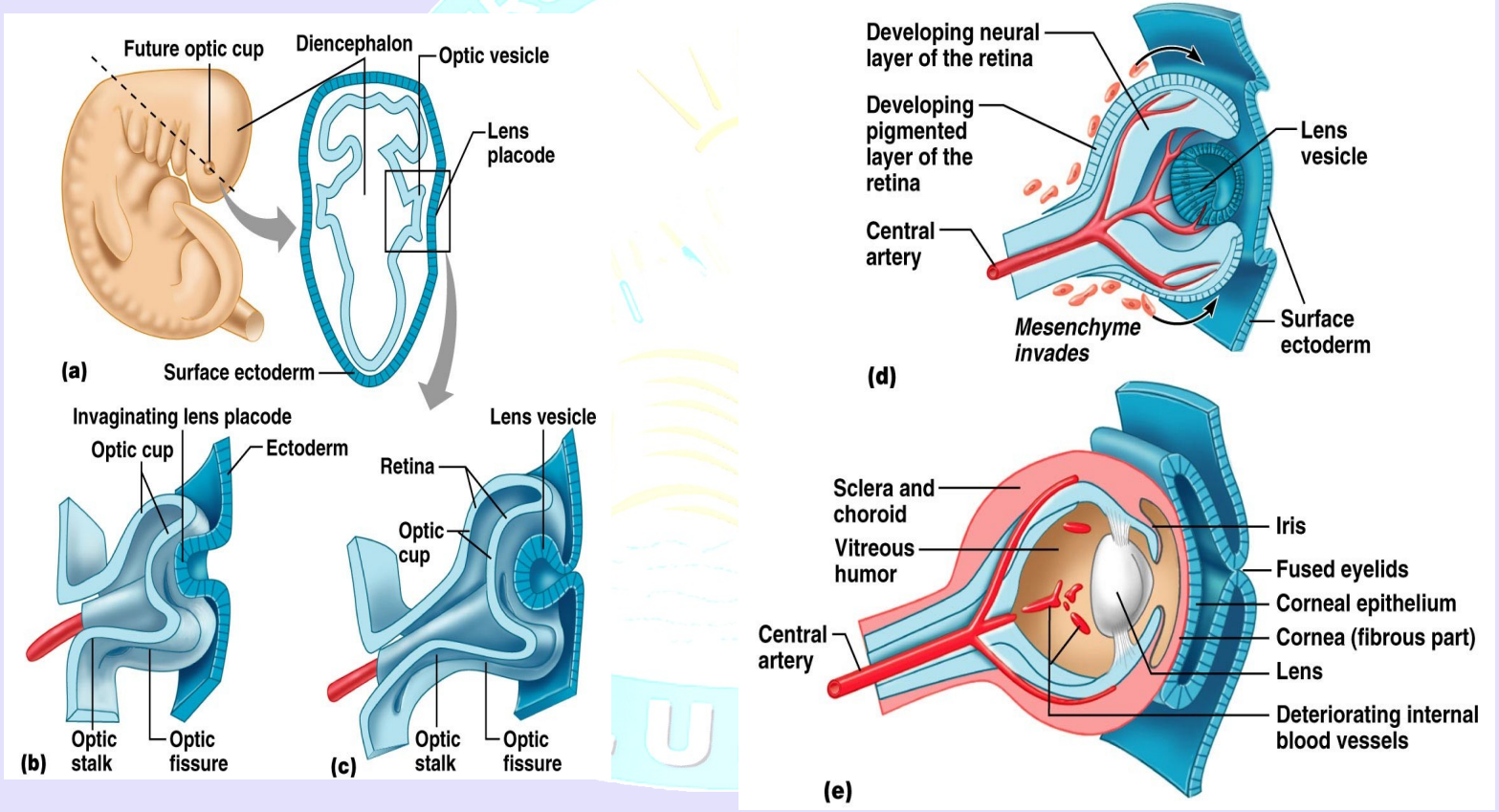
Embryonic Development of the Eye

Developing Retina

Developing Lens



Embryonic Development of the Ey



Embryonic Development of the Eye

- Invagination is not restricted to the central portion of the cup but also involves a part of the inferior surface *that* ***forms*** **the choroid fissure.**
- Formation of this fissure allows the hyaloid artery to reach the inner chamber of the eye.
- During the **7th week**,the lips of the choroid fissure fuse,and the mouth of the optic cup becomes a round opening, the future **pupil**.
- During these events, cells of the surface ectoderm, initially in contact with the optic vesicle, begin to elongate and form the **lens placode.**
- This placode subsequently invaginates and develops into the **lens vesicle**.

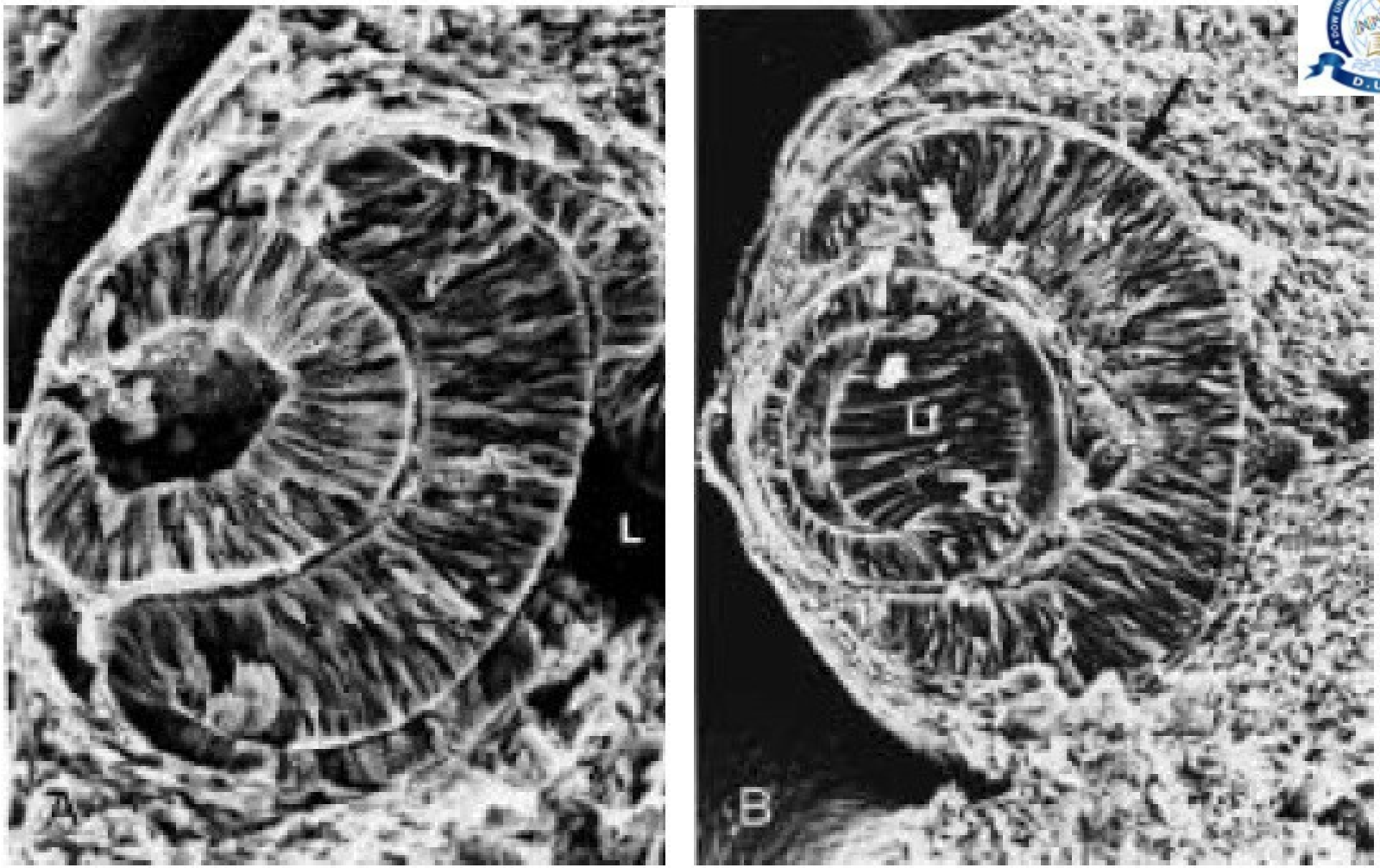
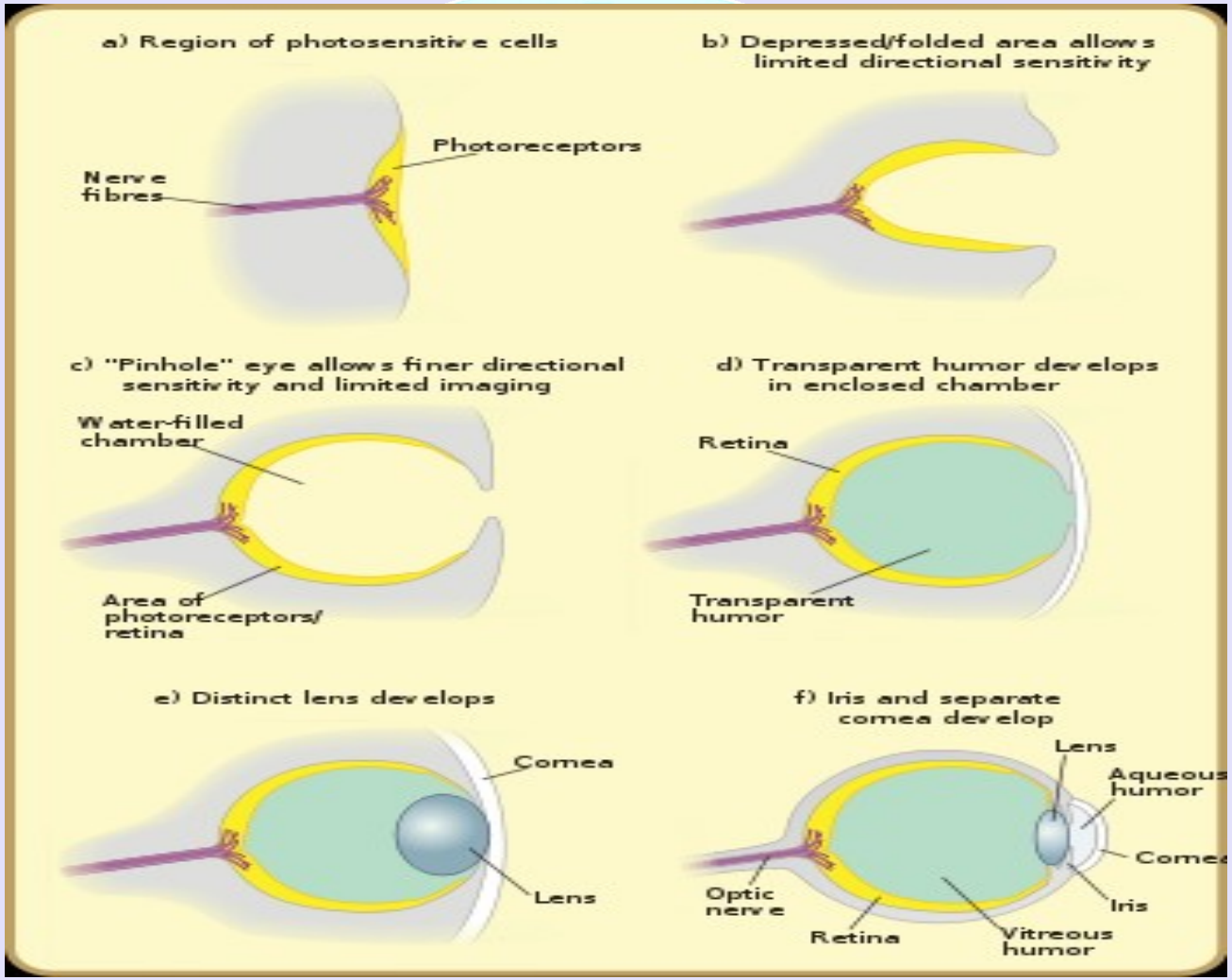


Figure 17.4 Scanning electron micrographs of sections through the eyes of mouse embryos at stages equivalent to **(A)** 6 weeks and **(B)** 7 weeks in the human. **A.** The forming lens vesicle, not entirely closed; the two layers of the optic cup; and the lumen (*L*) of the optic stalk. (Compare with Fig. 17.2**C**.) **B.** Lens fibers (*Lf*) and neural (*N*) and pigment layers (*arrow*) forming. (Compare with Fig. 17.3.)

Embryonic Development of the Eye



Retina, Iris, and Ciliary Body

- The outer layer of optic cup, characterized by small pigment granules is known as **pigmented layer of retina.**
- Development of the inner **(neural) layer of the optic cup is more complicated.**
- The posterior four-fifths,the **pars optica retinae, contains cells bordering intraretinal** space that differentiate into light-receptive elements, **rods & cones.**
- On the surface is a fibrous layer that contains axons of nerve cells of the deeper layers. Nerve fibers in this zone converge toward the optic stalk, which develops into the optic nerve. Hence, light impulses pass through most layers of the retina before they reach the rods and cones.

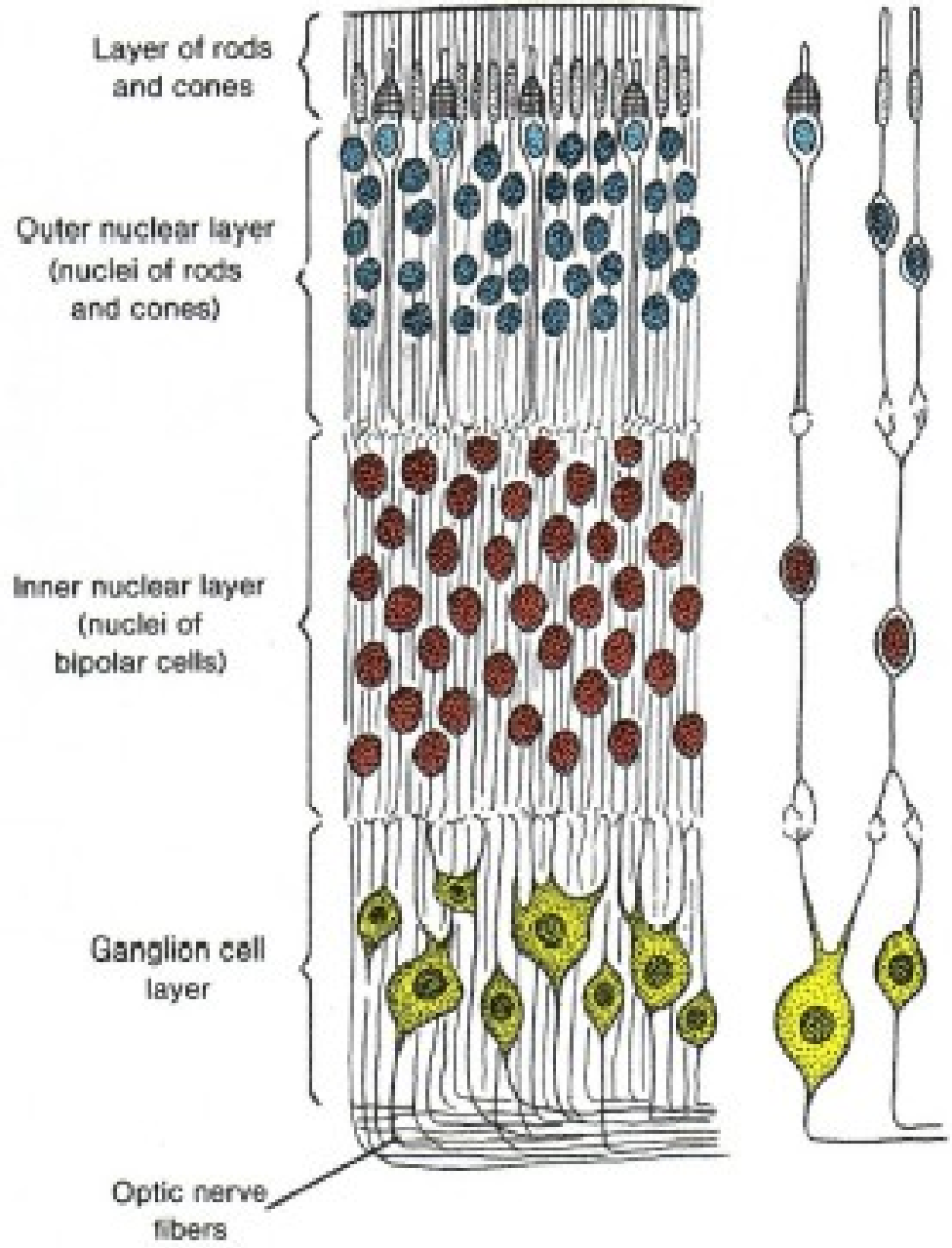


Figure 17.5. Various layers of the pars optica retinae in a fetus of approximately 25 weeks.

Retina, Iris, and Ciliary Bod

- The anterior fifth of the inner layer, the **pars ceca retinae, remains one cell** layer thick.
- It later divides into the **pars iridica retinae, which forms the inner** layer of the iris, and the **pars ciliaris retinae, which participates in formation** of the **ciliary body.**
- Meanwhile, the region between the optic cup and the overlying surface epithelium is filled with loose mesenchyme.
- The **sphincter and dilator pupillae muscles form in this tissue.**
- **These** muscles develop from underlying ectoderm of optic cup.
- The **pars ciliaris retinae is easily recognized by its marked folding** *Externally it is covered by a layer of mesenchyme that* forms the **ciliary muscle; on the inside it is connected to the lens by a network** of elastic fibers, the **suspensory ligament or zonula .**

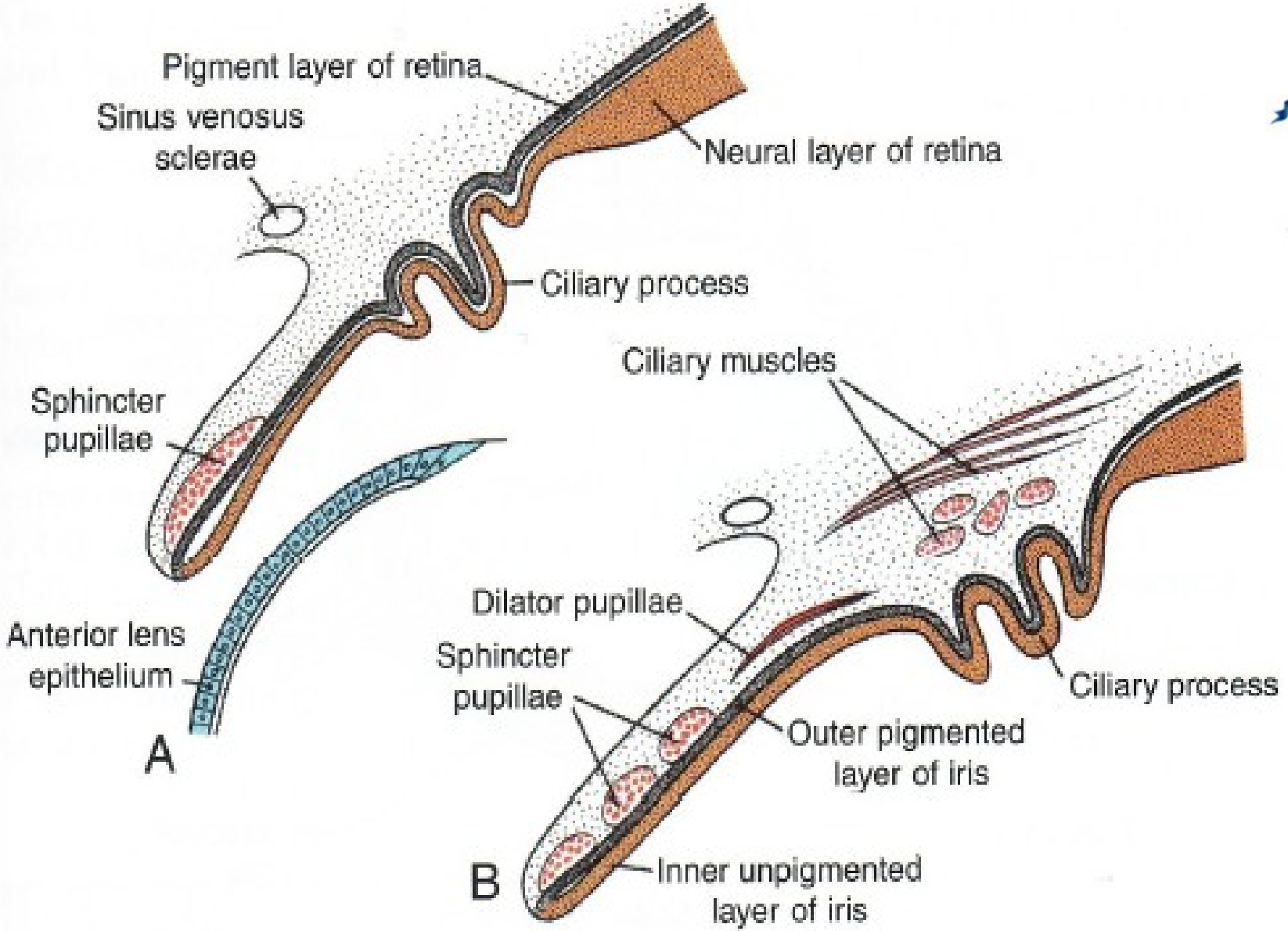


Figure 17.6. Development of the iris and ciliary body. The rim of the optic cup is covered by mesenchyme, in which the sphincter and dilator pupillae develop from the underlying ectoderm.

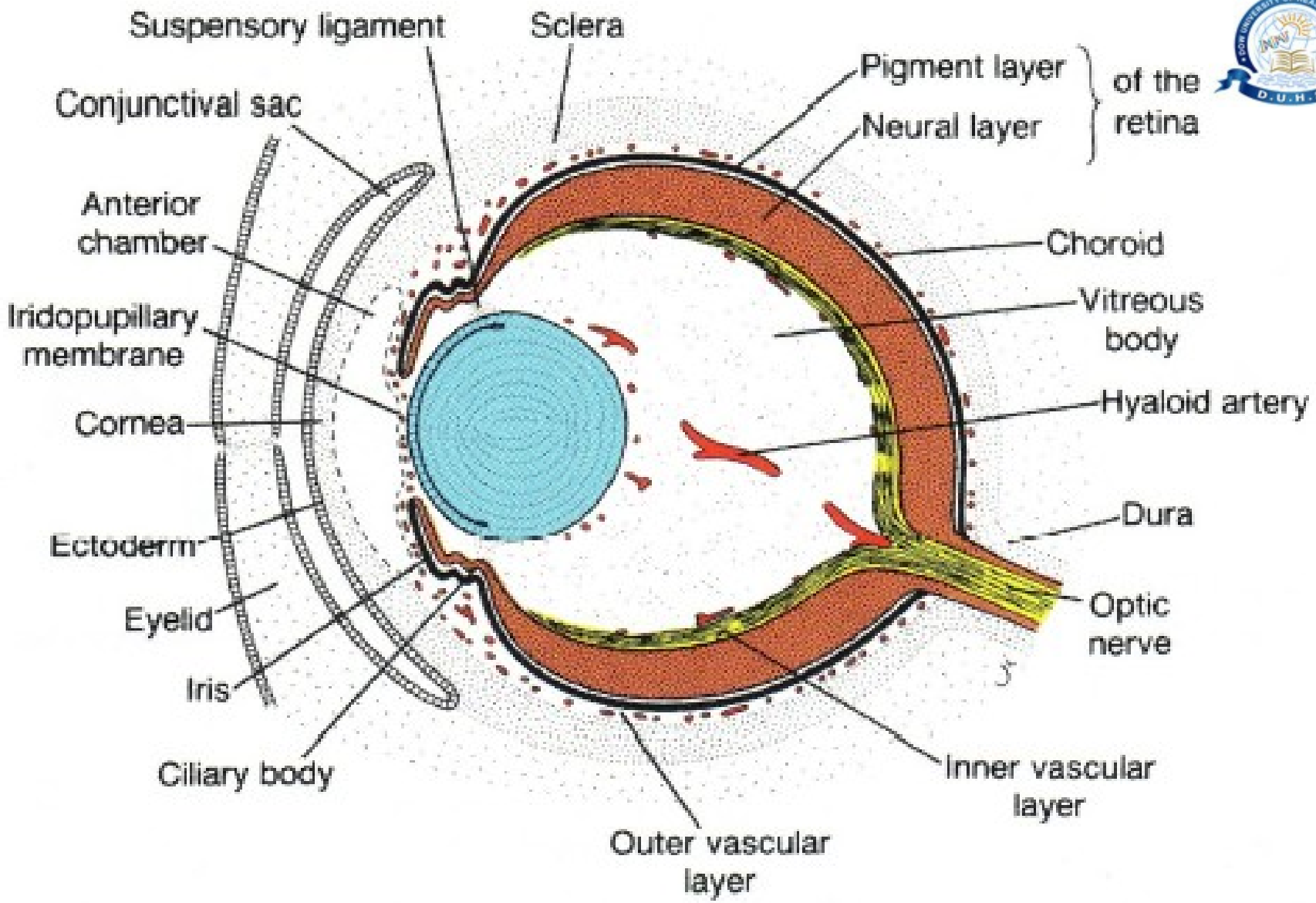


Figure 17.7. Section through the eye of a 15-week fetus showing the anterior chamber, iridopupillary membrane, inner and outer vascular layers, choroid, and sclera.

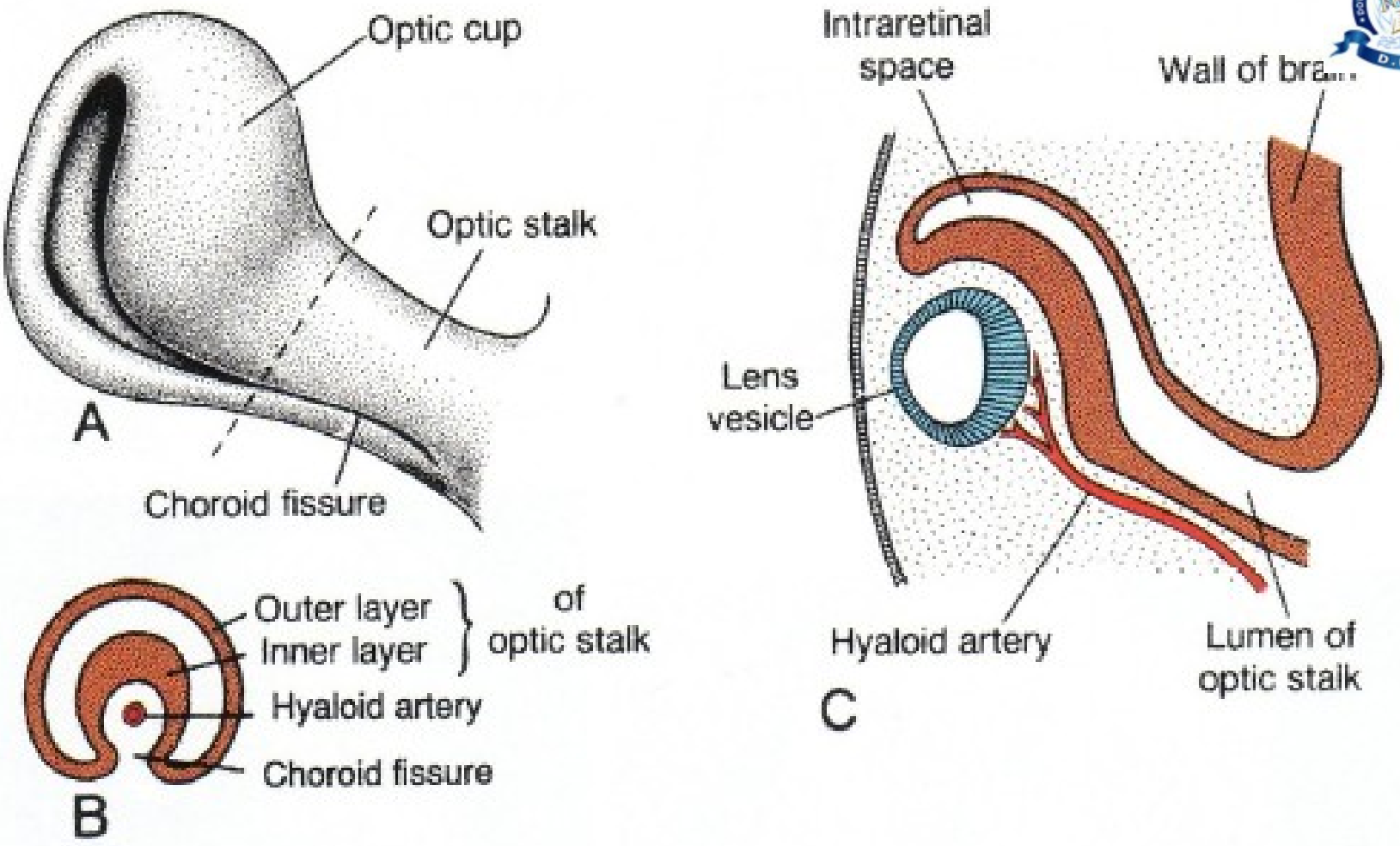


Figure 17.2. **A.** Ventrolateral view of the optic cup and optic stalk of a 6-week embryo. The choroid fissure on the undersurface of the optic stalk gradually tapers off. **B.** Transverse section through the optic stalk as indicated in **A**, showing the hyaloid artery in the choroid fissure. **C.** Section through the lens vesicle, the optic cup, and optic stalk at the plane of the choroid fissure.

Lens

- Shortly after formation of the lens vesicle, *cells of the posterior wall* begin to elongate anteriorly and form long fibers that gradually fill the lumen of the vesicle
- *By the end of the seventh week, these* **primary lens fibers reach the anterior wall of the lens vesicle.**
- **Growth of the** lens is not finished at this stage, however, since new (secondary) lens fibers are continuously added to the central core.

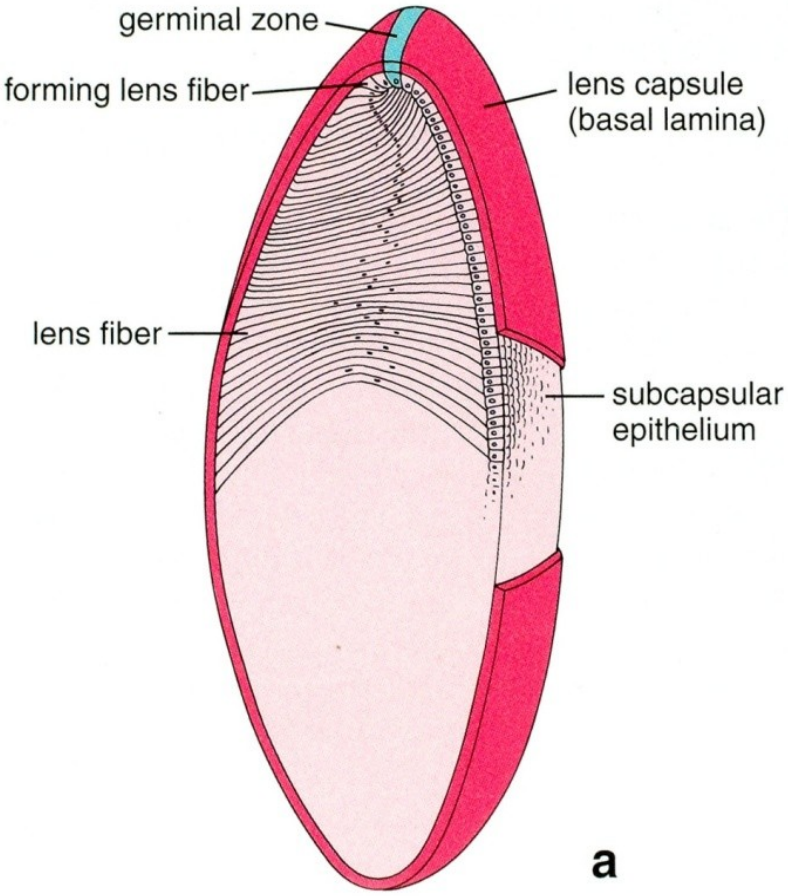


FIGURE 23.15
Structure of the lens. a. This schematic drawing of the lens indicates its structural components. Note that the capsule of the lens is formed by the basal lamina of the lens fibers and the subcapsular epithelium located on the anterior surface of the lens. Also note the location of the germinal zone at the equatorial area of the lens. **b.** This high-

Choroid, Sclera, and Cornea

- At the end of the fifth week, the eye primordium is completely surrounded by loose mesenchyme.
- This tissue soon differentiates into an inner layer comparable with the pia mater of the brain and an outer layer comparable with the dura mater.
- The inner layer later forms a highly vascularized pigmented layer known as the **choroid;**
- **the outer layer develops into the sclera and is** continuous with the dura mater around the optic nerve.

Choroid, Sclera, and Cornea

- The **anterior chamber forms through vacuolization and splits** the mesenchyme into an inner layer in front of the lens and iris, the **iridopupillary membrane, and an outer layer continuous with the sclera, the substantia propria of the cornea .**
- **The anterior chamber itself is lined by flattened** mesenchymal cells.
- Hence, the cornea is formed by

(a) *an epithelial layer* derived from the surface ectoderm,
(b) *the* ***substantia propria or stroma, which*** is continuous with the sclera, and
(c) *an epithelial layer, which borders the anterior* chamber.

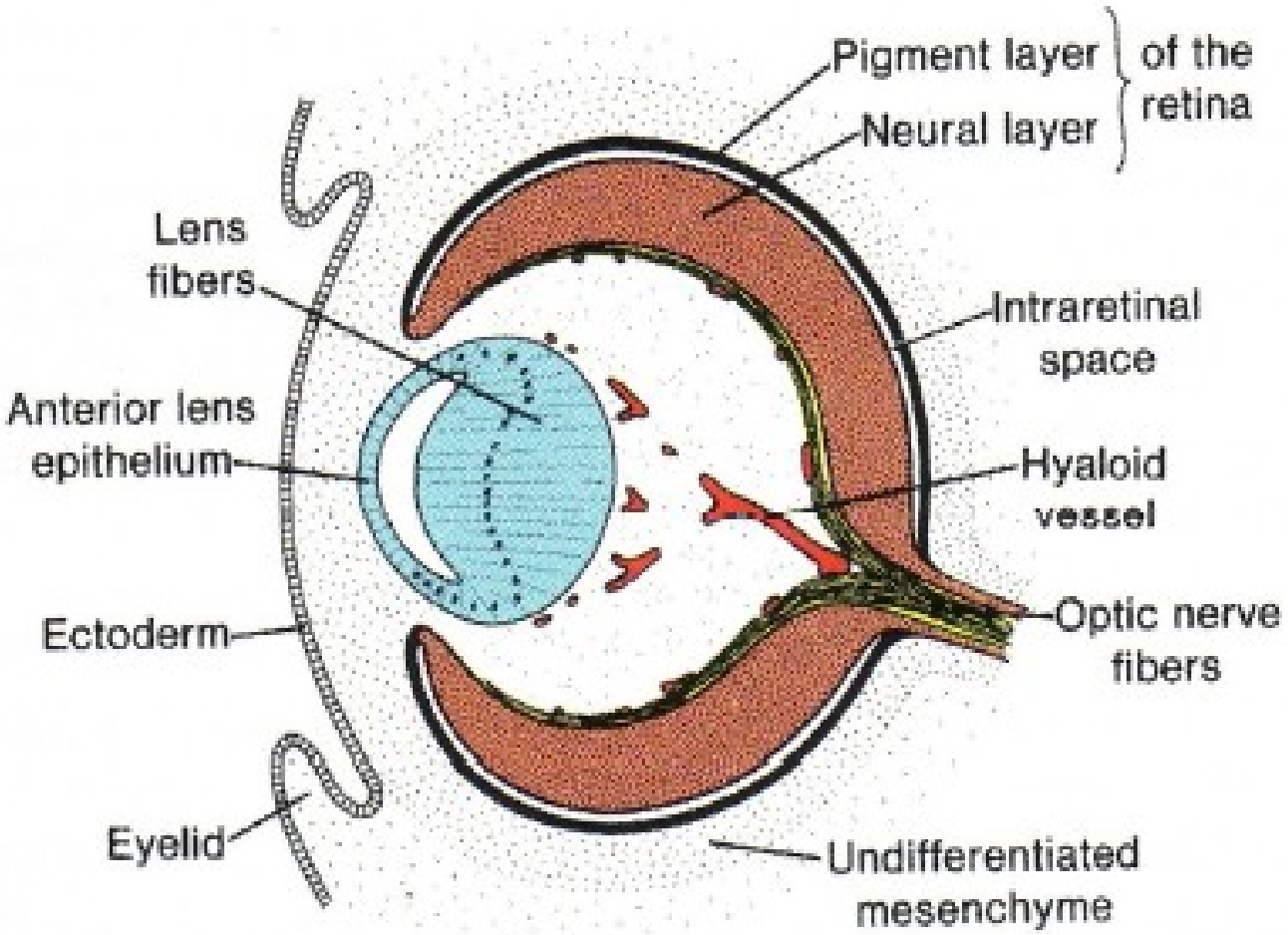


Figure 17.3. Section through the eye of a 7-week embryo. The eye primordium is completely embedded in mesenchyme. Fibers of the neural retina converge toward the optic nerve.

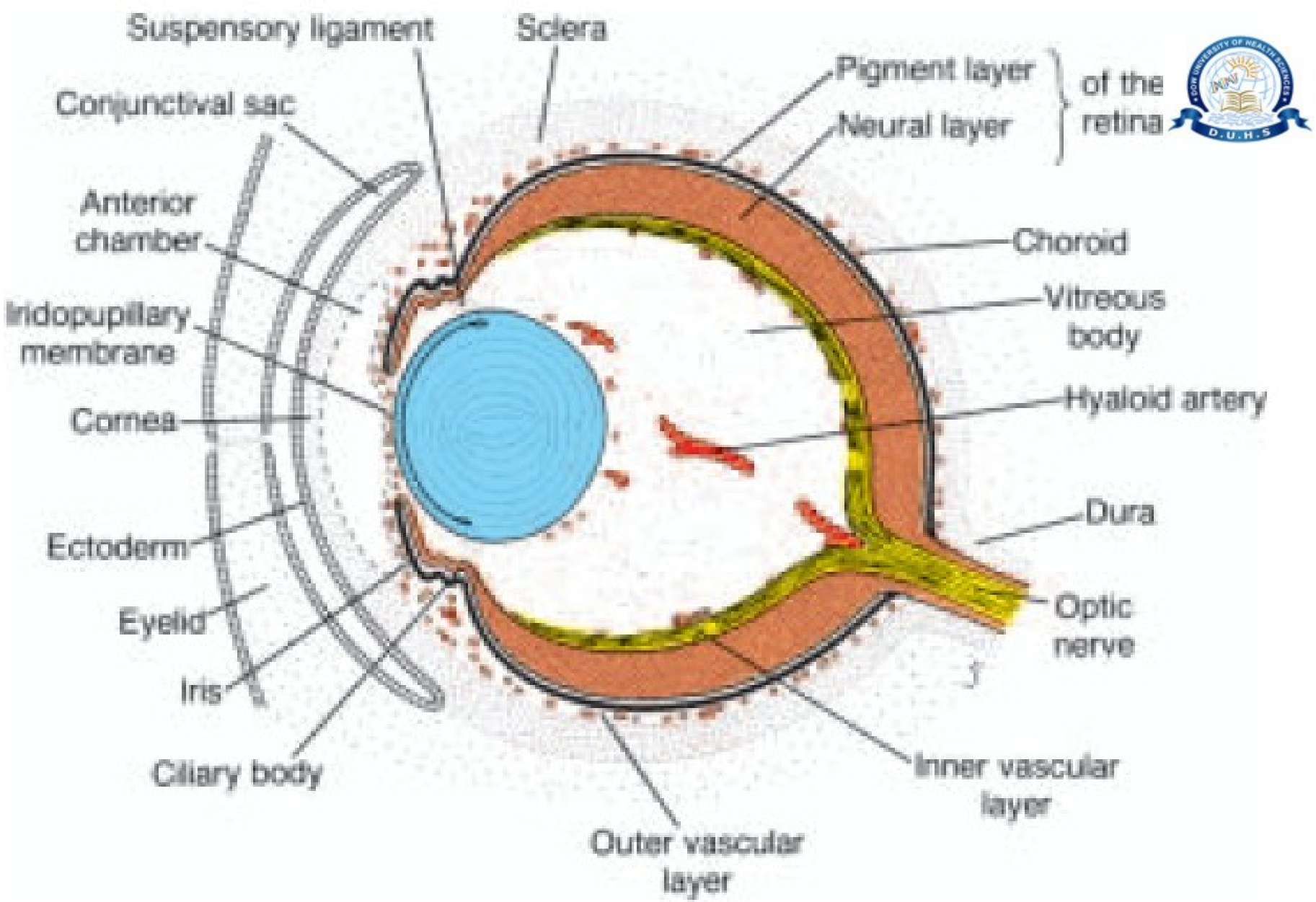


Figure 17.7 Section through the eye of a 15-week fetus showing the anterior chamber, iridopupillary membrane, inner and outer vascular layers, choroid, and sclera.

Vitreous Body

- Mesenchyme not only surrounds the eye primordium from the outside but also invades the inside of the optic cup by way of the choroid fissure.
- Here it forms the hyaloid vessels, which during intrauterine life supply the lens and form the vascular layer on the inner surface of the retina.
- The interstitial spaces fill with a transparent gelatinous substance, forming the **vitreous body. The hyaloid vessels in this region are obliterated and** disappear during fetal life, leaving behind the hyaloid canal.

Optic Nerve

The optic cup is connected to the brain by the optic stalk, which has a groove, the **choroid fissure, on its ventral surface.**

In this groove are the hyaloid vessels.

The nerve fibers of the retina returning to the brain lie among cells of the inner wall of the stalk.

During the seventh week, the choroid fissure closes, and a narrow tunnel forms inside the optic stalk.

As a result of the continuously increasing number of nerve fibers, the inner wall of the stalk grows, and the inside and outside walls of the stalk fuse,

Cells of the inner layer provide a network of neuroglia that support the optic nerve fibers.

The optic stalk is thus transformed into the **optic nerve.**

Its center contains a portion of the hyaloid artery, later called the **central artery of the retina.**

On the outside, a continuation of the choroid and sclera, the **pia arachnoid and dura layer of the nerve, respectively, surround the optic nerve.**

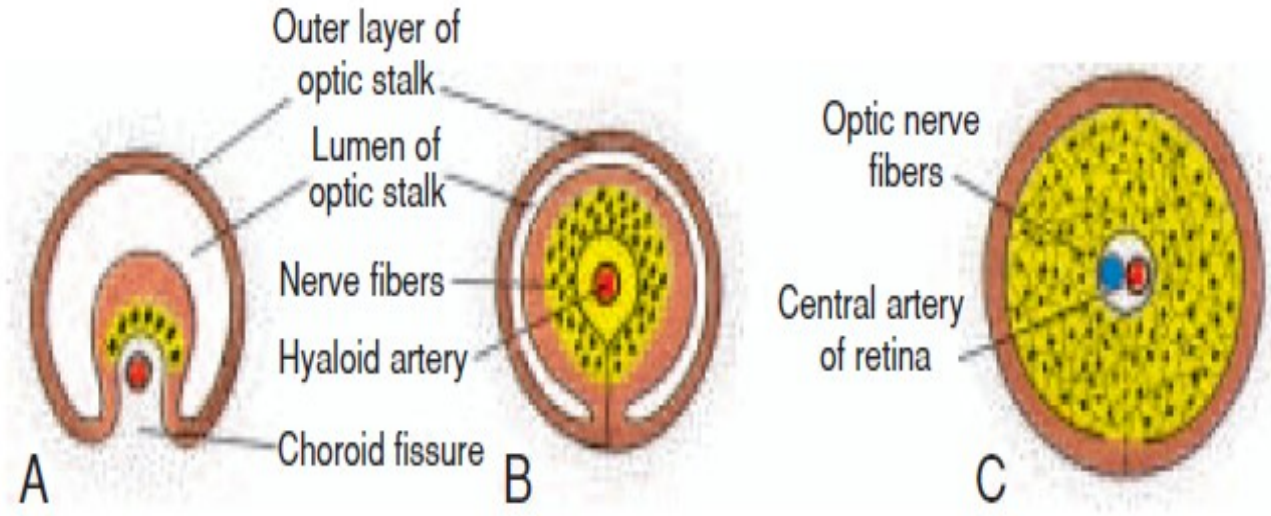


Figure 17.8 Transformation of the optic stalk into the optic nerve. **A.** Sixth week (9 mm). **B.** Seventh week (15 mm). **C.** Ninth week. Note the central artery of the retina in the optic nerve.

Molecular Regulation of Eye Development

- ***PAX6 is the key regulatory gene for eye development.***
- ***It is a member of the PAX (paired box) family of transcription factors and contains two DNA binding*** motifs that include a paired domain and a paired type homeodomain.
- *PROX1, is upregulated by PAX6, while BMP-7 expression in the lens ectoderm* is increased to maintain expression of *SOX2 and PAX6. Finally, the combined* expression of *PAX6, SOX2, and LMAF initiates expression of genes responsible* for formation of lens crystalline proteins, while *PROX1 expression regulates* genes controlling cell proliferation

Eye Defects at birth

Aniridia

- Partial or complete absence of the iris of the eye. This rare condition, usually present at birth, results in impaired vision and sensitivity to light.
- **Anopthalmia**

Rare condition in which one or both eyes do not form during pregnancy. When both eyes are affected, blindness results.

AstigmatismCommon vision condition, usually present from birth, caused by an irregularly curved cornea or lens.

Eye defects at birth

- **Cataract**

A condition in which the lens of the eye, which is normally clear, becomes cloudy or opaque.

- **Coloboma**

A cleft or gap in some part of the eye, such as the iris, lens, or retina, that is caused by a defect in the development of the eyeball.

- **Retinitis pigmentosa**

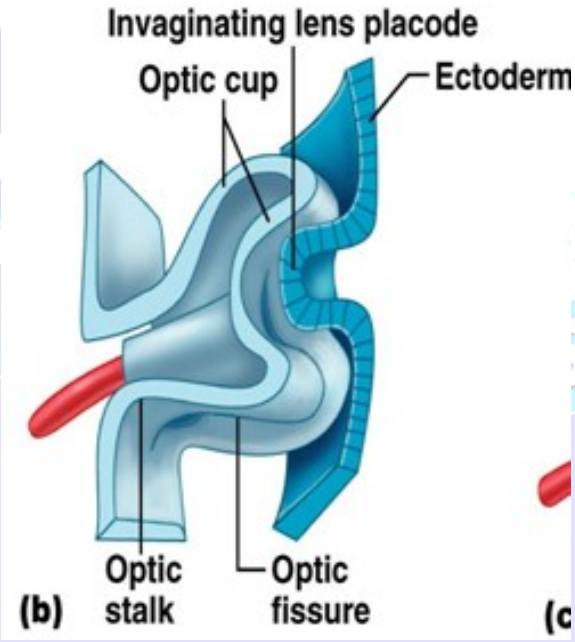
Degeneration of the retina, resulting in decreased night vision, a gradual loss of peripheral vision, and in some cases, loss of central vision.

- **Retinopathy of prematurity (ROP)**Condition associated with premature birth, in which the growth of normal blood vessels in the retina stops, and abnormal blood vessels develop

Self Assessment

Q1. How and when the development of eye in human starts?

Q2.Give the adult derivatives of The structures shown in the diagram below.



Thank You